Algebra 2 — Practice Worksheet

Sequences · Functions · Systems · Rational Expressions

Name: _____ Date: _____

How this packet is organized. Seven topics grouped into four parts. Each topic has a short **Summary**, one or two worked **Examples**, and **5 practice problems** rising in difficulty: **(Easy)** (1) → **(Medium)** (2) → **(Hard)** (2). **Full worked solutions are in the Answer Key at the end.**

Part I — Sequences: Topic 1. **Part II — Functions:** Topic 2. **Part III — Systems:** Topics 3–4. **Part IV — Rational & Radical Expressions and Equations:** Topics 5–7.

Topic 1 Arithmetic Sequences

An **arithmetic sequence** is a list of numbers with a **constant common difference** d between consecutive terms.

$$d = a_n - a_{n-1} \qquad \text{(same gap every step)}$$

The **n-th term formula** lets you jump straight to any term without listing them all:

$$\boxed{a_n = a_1 + (n-1)\,d}$$

where a_1 is the first term and n is the term number. **Two useful moves:** (1) find a specific term by plugging in n; (2) find d and a_1 from two known terms, then build the formula.

Example 1 (find a term). If $a_1 = 3$ and $d = 5$, find a_{10}.
$a_{10} = 3 + (10-1)(5) = 3 + 45 = 48$.

Example 2 (build the formula). For 7, 11, 15, 19, …: $d = 4$, $a_1 = 7$, so $a_n = 7 + (n-1)(4) = 4n + 3$. Then $a_{20} = 4(20) + 3 = 83$.

Example 3 (from two terms). If $a_4 = 14$ and $a_9 = 34$: subtract to get $a_9 - a_4 = 5d \Rightarrow 20 = 5d \Rightarrow d = 4$; then $a_1 = a_4 - 3d = 14 - 12 = 2$, so $a_n = 2 + (n-1)(4) = 4n - 2$.

Practice

1. **(Easy)** Given $a_1 = 5$ and $d = 3$, find a_8.

2. **(Medium)** Write the n-th term formula for 12, 9, 6, 3, … and then find a_{15}.

3. **(Medium)** In an arithmetic sequence $a_1 = -4$ and $a_{12} = 29$. Find the common difference d and the value of a_{20}.

4. **(Hard)** The 5th term of a sequence is 22 and the 11th term is 52. Write the n-th term formula and find a_{30}.

5. **(Hard)** How many terms are in the arithmetic sequence 4, 10, 16, …, 106?

Topic 2 Piecewise Functions

A **piecewise function** uses **different rules on different parts of the domain**. For example

$$f(x) = \begin{cases} 2x+1, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

To evaluate: decide which interval the input falls in, then use that piece's rule.
To graph: graph each piece *only* over its interval. Use an **open circle** $\circ$ at an endpoint with a strict inequality ($<$ or $>$) and a **closed dot** $\bullet$ with $\leq$ or $\geq$.
To write a piecewise rule from a graph: find each straight piece's equation ($y = mx + b$) and the x-interval it covers; note which endpoints are open vs. closed.

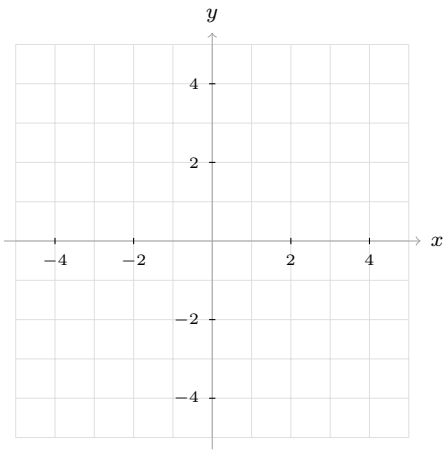
Example (evaluate). For the f above: $f(-2) = 2(-2)+1 = -3$ (used the top piece since $-2 < 0$); $f(3) = 3^2 = 9$ (bottom piece since $3 \geq 0$).

Example (graph). $g(x) = \begin{cases} -x+2, & x \leq 1 \\ 3, & x > 1 \end{cases}$ is a line of slope -1 up to a *closed* dot at $(1, 1)$, then a horizontal ray at height 3 starting with an *open* circle at $(1, 3)$.

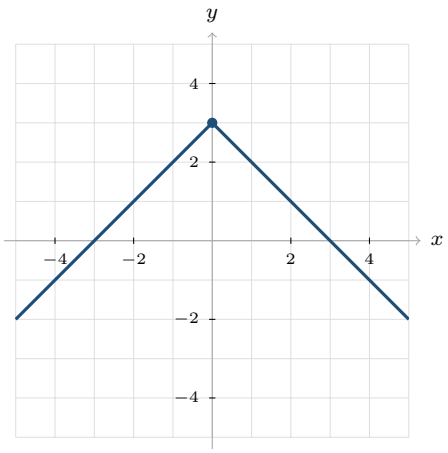
Practice

1. (Easy) Let $f(x) = \begin{cases} x+4, & x < 2 \\ 2x, & x \geq 2 \end{cases}$ Find $f(-1)$, $f(2)$, and $f(5)$.

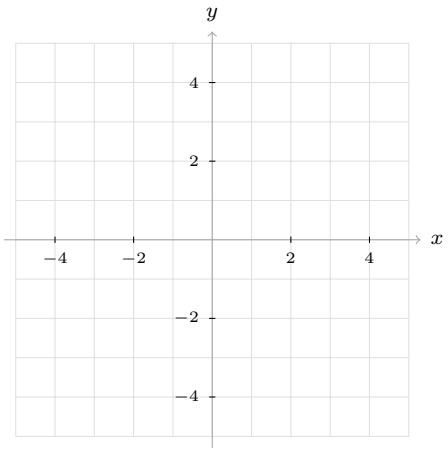
2. (Medium) Graph $f(x) = \begin{cases} x+1, & x < 1 \\ 4-x, & x \geq 1 \end{cases}$ on the grid. Mark open/closed endpoints.



3. (Medium) Write the piecewise function shown in the graph.



4. **(Hard)** Graph $f(x) = \begin{cases} x+2, & x < 0 \\ 1, & 0 \le x \le 2 \\ -x+4, & x > 2 \end{cases}$ and state the **range**.



5. **(Hard)** A rideshare charges a \$4 base fare plus \$2 per mile for the first 5 miles, then \$1.50 per mile after that. Write a piecewise function $C(x)$ for the total cost of an x-mile trip, then find $C(3)$ and $C(8)$.

Topic 3 Systems of Linear Equations

A **system** is two equations that must be true at the same time; the **solution** (x, y) is the point that satisfies both. Three algebraic tools:

- **Substitution:** solve one equation for a variable, then substitute into the other.
- **Elimination:** add or subtract multiples of the equations to cancel one variable.
- **Graphing:** the intersection point is the solution.

Outcomes: one solution (lines cross), **no solution** (parallel lines), or **infinitely many** (same line).

Example (substitution). $y = 2x - 1$ and $3x + y = 9$. Substitute: $3x + (2x - 1) = 9 \Rightarrow 5x = 10 \Rightarrow x = 2$, so $y = 3$. Solution $(2, 3)$.

Example (elimination). $2x + 3y = 12$ and $2x - y = 4$. Subtract the equations: $4y = 8 \Rightarrow y = 2$; back-substitute $2x - 2 = 4 \Rightarrow x = 3$. Solution $(3, 2)$.

Practice

1. **(Easy)** Solve by substitution: $y = x + 2, \quad 3x + y = 10$.

2. **(Medium)** Solve by elimination: $3x + 2y = 16, \quad x - 2y = -8$.

3. **(Medium)** Solve: $4x + 3y = 10, \quad 2x - y = 0$.

4. **(Hard)** Solve: $3x + 4y = -1, \quad 5x + 6y = 1$.

5. **(Hard)** A theater sells adult tickets for \$12 and child tickets for \$8. One night they sold 200 tickets and collected \$2080. How many of each type were sold? (Set up and solve a system.)

Topic 4 Graphing Systems of Linear Inequalities

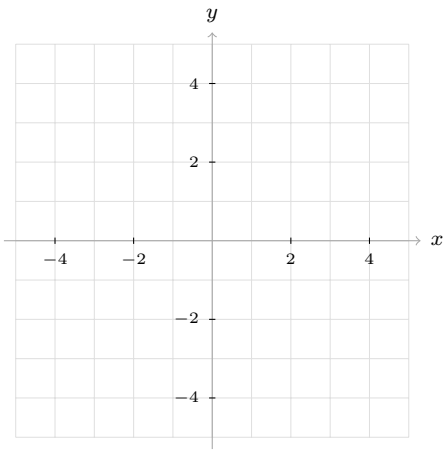
To graph a system of inequalities, graph each one and take the **overlap**:

- Draw the boundary line: **solid** for $\leq$ or $\geq$, **dashed** for $<$ or $>$.
- Shade the correct side. **Test a point** not on the line (often the origin): if it makes the inequality true, shade that side; if false, shade the other side.
- The **solution set** is the region where *all* shadings overlap.

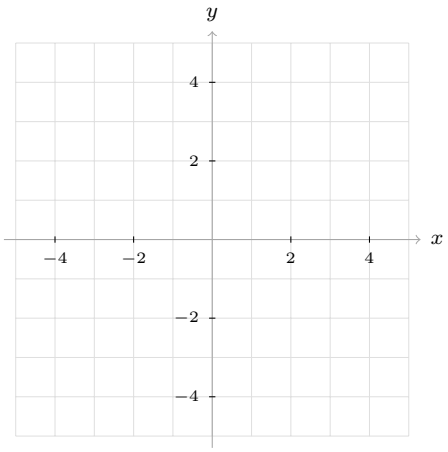
Example. For $y \geq x - 1$ (solid line, shade above) and $y < -x + 3$ (dashed line, shade below), the solution is the wedge where the two shaded regions overlap. Testing $(0, 0)$: $0 \geq -1$ ✓ and $0 < 3$ ✓, so $(0, 0)$ lies in the solution.

Practice (Graph each system; shade the solution region.)

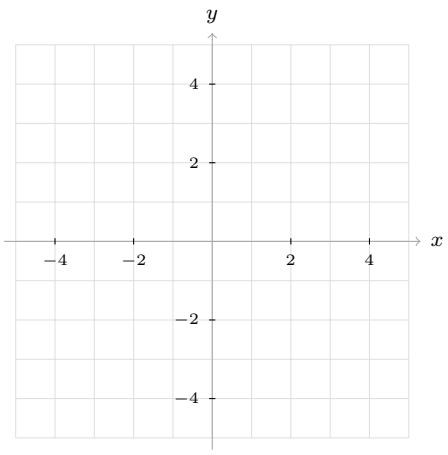
1. **(Easy)** $y \geq 1$ and $x \leq 3$.



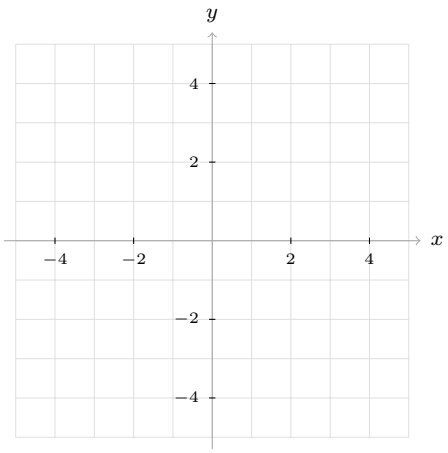
2. **(Medium)** $y < x + 2$ and $y \geq -x$.



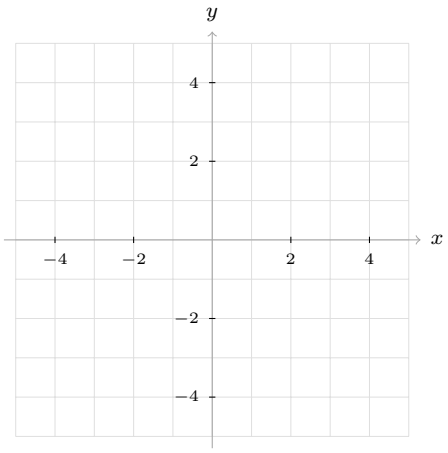
3. **(Medium)** $y \geq 2x - 3$ and $y < -x + 4$.



4. **(Hard)** $y \leq -x + 5$, $y \geq x - 1$, and $x \geq 0$ (three inequalities).



5. **(Hard)** $2x + y < 4$ and $y \geq \frac{1}{2}x - 2$. Then state whether $(0, 0)$ is a solution.



Topic 5 Evaluating Expressions with Rational Exponents

A **rational exponent** means a root. The denominator is the root; the numerator is the power:

$$a^{1/n} = \sqrt[n]{a}, \qquad \boxed{a^{m/n} = \left(\sqrt[n]{a}\right)^m = \sqrt[n]{a^m}}$$

A **negative** rational exponent means take the reciprocal:

$$a^{-m/n} = \frac{1}{a^{m/n}}.$$

Best strategy for numbers: take the *root first* (numbers stay small), then apply the power.

Example 1. $27^{1/3} = \sqrt[3]{27} = 3$.

Example 2. $16^{3/4} = \left(16^{1/4}\right)^3 = 2^3 = 8$.

Example 3 (negative). $8^{-2/3} = \dfrac{1}{8^{2/3}} = \dfrac{1}{\left(8^{1/3}\right)^2} = \dfrac{1}{2^2} = \dfrac{1}{4}$.

Practice (Evaluate; give exact values.)

1. **(Easy)** $25^{1/2}$

2. **(Medium)** $8^{4/3}$

3. **(Medium)** $81^{3/4}$

4. **(Hard)** $32^{-3/5}$

5. **(Hard)** $\left(\frac{27}{8}\right)^{-2/3}$

Topic 6 Solving Rational Equations

A **rational equation** has one or more variables in a denominator. To solve:

1. Find the **LCD** (least common denominator) of all fractions.
2. Multiply *every* term by the LCD to clear the denominators.
3. Solve the resulting equation (it may be linear or quadratic).
4. **Check for extraneous solutions:** any answer that makes a denominator 0 is rejected.

For a simple proportion $\frac{A}{B} = \frac{C}{D}$, you may **cross-multiply**: $AD = BC$.

Example 1. $\frac{x}{3} + \frac{1}{2} = \frac{5}{6}$. Multiply every term by 6: $2x + 3 = 5 \Rightarrow 2x = 2 \Rightarrow x = 1$.

Example 2 (extraneous). $\frac{x}{x-3} = \frac{3}{x-3} + 4$. Multiply by $(x-3)$: $x = 3 + 4(x-3) \Rightarrow x = 3$. But $x = 3$ makes a denominator 0, so it is **rejected**: *no solution.*

Practice

1. **(Easy)** $\frac{x}{4} + \frac{1}{2} = \frac{5}{4}$

2. **(Medium)** $\frac{5}{x} = \frac{10}{x+3}$

3. **(Medium)** $\frac{2}{x-1} + 3 = \frac{5}{x-1}$

4. **(Hard)** $\frac{12}{x} = x + 4$

5. **(Hard)** $\frac{x}{x-3} = \frac{3}{x-3} + 4$

Topic 7 Solving Radical (Root) Equations

A **radical equation** has a variable **under a radical** (usually a square root). To solve:

1. **Isolate the radical** on one side of the equation.
2. **Undo the root** by raising both sides to the matching power (square a square root, cube a cube root).
3. Solve the resulting equation (it may be linear or quadratic).
4. **Check every answer in the original equation.** Squaring can create *extraneous* solutions — values that satisfy the squared equation but not the original one.

Key fact: $(\sqrt{a})^2 = a$, so squaring removes a square root — but only *after* it is alone.

Example 1. $\sqrt{x+3} = 4$. Square both sides: $x + 3 = 16 \Rightarrow x = 13$. Check: $\sqrt{16} = 4$ ✓.

Example 2 (isolate first). $\sqrt{2x-1}+3 = 8$. Isolate: $\sqrt{2x-1} = 5$; square: $2x-1 = 25 \Rightarrow x = 13$.

Example 3 (extraneous). $\sqrt{x+2} = x$. Square: $x+2 = x^2 \Rightarrow x^2-x-2 = 0 \Rightarrow (x-2)(x+1) = 0$. Candidates $x = 2, -1$. Check: $x = 2$ gives $\sqrt{4} = 2$ ✓; $x = -1$ gives $\sqrt{1} = 1 \neq -1$ **(reject)**. Only $x = 2$.

Practice

1. **(Easy)** $\sqrt{x} = 7$

2. **(Medium)** $\sqrt{x-4} = 6$

3. **(Medium)** $\sqrt{3x+1} - 2 = 3$

4. **(Hard)** $\sqrt{x+6} = x$

5. **(Hard)** The time (in seconds) for an object to fall d feet is modeled by $t = \sqrt{\dfrac{d}{16}}$. If a stone falls for 2.5 seconds before it hits the ground, how far did it fall?

Answer Key

Full worked solutions

Topic 1 — Arithmetic Sequences

1. $a_8 = a_1 + 7d = 5 + 7(3) = \boxed{26}$.

2. $d = 9 - 12 = -3$, $a_1 = 12$, so $a_n = 12 + (n-1)(-3) = \boxed{-3n + 15}$. Then $a_{15} = -3(15) + 15 = \boxed{-30}$.

3. $a_{12} = a_1 + 11d \Rightarrow -4 + 11d = 29 \Rightarrow 11d = 33 \Rightarrow \boxed{d = 3}$. Then $a_{20} = -4 + 19(3) = \boxed{53}$.

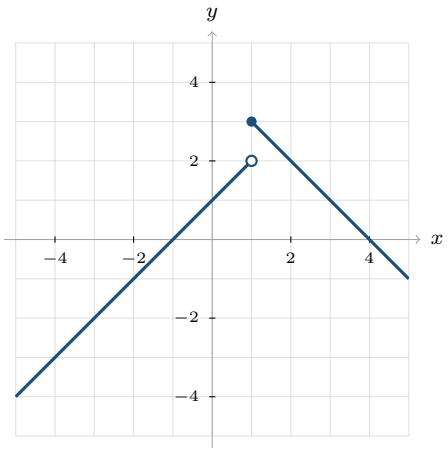
4. $a_{11} - a_5 = 6d \Rightarrow 52 - 22 = 30 \Rightarrow d = 5$. $a_1 = a_5 - 4d = 22 - 20 = 2$, so $a_n = 2 + (n-1)(5) = \boxed{5n - 3}$. Then $a_{30} = 5(30) - 3 = \boxed{147}$.

5. Here $a_1 = 4$, $d = 6$, so $a_n = 4 + (n-1)(6) = 6n - 2$. Set $6n - 2 = 106 \Rightarrow 6n = 108 \Rightarrow n = \boxed{18}$ terms.

Topic 2 — Piecewise Functions

1. $f(-1) = (-1) + 4 = \boxed{3}$ (since $-1 < 2$); $f(2) = 2(2) = \boxed{4}$ (since $2 \geq 2$); $f(5) = 2(5) = \boxed{10}$.

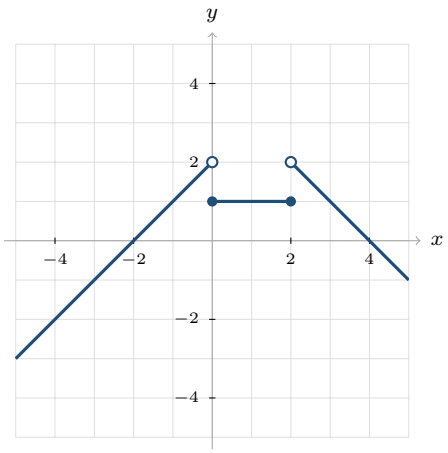
2. Line $y = x + 1$ for $x < 1$ ends at an *open* circle $(1, 2)$; ray $y = 4 - x$ for $x \geq 1$ starts at a *closed* dot $(1, 3)$ — a jump at $x = 1$.



3. Left piece has slope $+1$ through $(0, 3)$: $y = x + 3$ for $x \leq 0$. Right piece has slope -1 through $(0, 3)$: $y = -x + 3$ for $x > 0$. So

$$f(x) = \begin{cases} x + 3, & x \leq 0 \\ -x + 3, & x > 0 \end{cases}$$

4. Endpoints: open $(0, 2)$ & closed $(0, 1)$ at $x = 0$; closed $(2, 1)$ & open $(2, 2)$ at $x = 2$. **Range:** $(-\infty, 2)$.



5. First 5 miles cost $4 + 2x$; at 5 miles that is \$14, and each extra mile adds \$1.50:

$$C(x) = \begin{cases} 4 + 2x, & 0 \le x \le 5 \\ 1.5x + 6.5, & x > 5 \end{cases}$$

$C(3) = 4 + 2(3) = \boxed{\$10}$; $C(8) = 1.5(8) + 6.5 = 12 + 6.5 = \boxed{\$18.50}$.

Topic 3 — Systems of Linear Equations

1. Substitute $y = x + 2$: $3x + (x + 2) = 10 \Rightarrow 4x = 8 \Rightarrow x = 2$, $y = 4$. Solution $\boxed{(2, 4)}$.

2. Add the equations: $(3x+2y)+(x-2y) = 16+(-8) \Rightarrow 4x = 8 \Rightarrow x = 2$. Then $2-2y = -8 \Rightarrow y = 5$. Solution $\boxed{(2, 5)}$.

3. From the second equation $y = 2x$. Substitute: $4x + 3(2x) = 10 \Rightarrow 10x = 10 \Rightarrow x = 1$, $y = 2$. Solution $\boxed{(1, 2)}$.

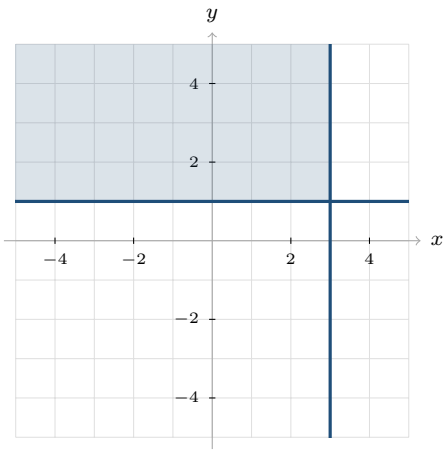
4. Multiply the first by 3 and the second by 2: $9x + 12y = -3$ and $10x + 12y = 2$. Subtract: $-x = -5 \Rightarrow x = 5$. Then $3(5) + 4y = -1 \Rightarrow 4y = -16 \Rightarrow y = -4$. Solution $\boxed{(5, -4)}$.

5. Let $a =$ adult, $c =$ child. $a+c = 200$ and $12a+8c = 2080$. Substitute $c = 200-a$: $12a+8(200-a) = 2080 \Rightarrow 4a = 480 \Rightarrow a = 120$. Then $c = 80$. $\boxed{120 \text{ adult},\ 80 \text{ child}}$.

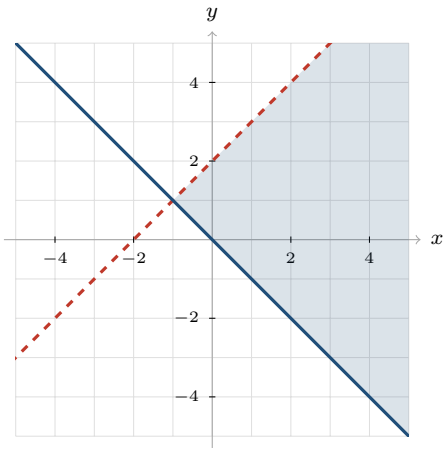
Topic 4 — Systems of Linear Inequalities

Shaded region = solution set; dashed = strict ($<, >$), solid = inclusive ($\le, \ge$).

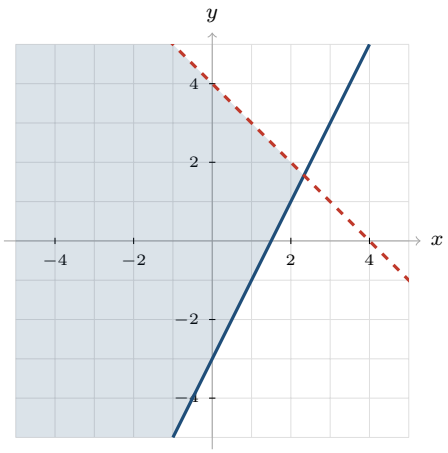
1. $y \ge 1$ (solid, shade up) and $x \le 3$ (solid, shade left). Overlap: the region at/above $y = 1$ and at/left of $x = 3$.



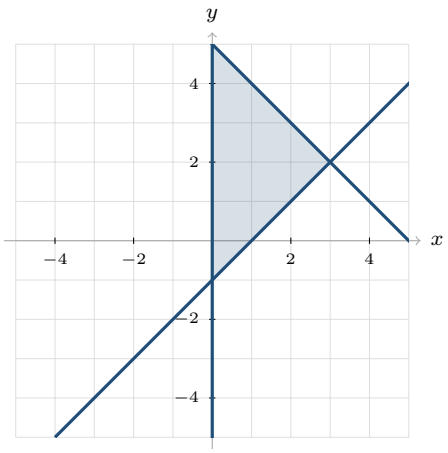
2. $y < x + 2$ (dashed, shade below) and $y \geq -x$ (solid, shade above); lines meet at $(-1, 1)$. Overlap opens to the right.



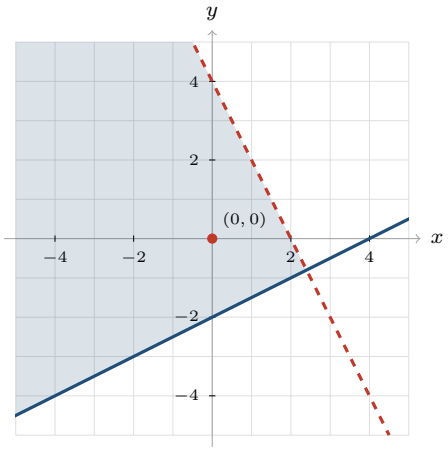
3. $y \geq 2x - 3$ (solid, shade above) and $y < -x + 4$ (dashed, shade below); lines meet at $\left(\frac{7}{3}, \frac{5}{3}\right)$. Overlap opens to the left.



4. $y \leq -x + 5$ (solid), $y \geq x - 1$ (solid), $x \geq 0$ (solid). Overlap is the triangle with vertices $(0, -1)$, $(0, 5)$, $(3, 2)$.



5. Rewrite the first as $y < -2x + 4$ (dashed, shade below); $y \geq \frac{1}{2}x - 2$ (solid, shade above); lines meet at $(2.4, -0.8)$. Test $(0,0)$: $2(0) + 0 = 0 < 4$ ✓ and $0 \geq -2$ ✓, so $(0,0)$ **IS a solution**.



Topic 5 — Rational Exponents

1. $25^{1/2} = \sqrt{25} = \boxed{5}$.

2. $8^{4/3} = \left(8^{1/3}\right)^4 = 2^4 = \boxed{16}$.

3. $81^{3/4} = \left(81^{1/4}\right)^3 = 3^3 = \boxed{27}$.

4. $32^{-3/5} = \dfrac{1}{32^{3/5}} = \dfrac{1}{\left(32^{1/5}\right)^3} = \dfrac{1}{2^3} = \boxed{\dfrac{1}{8}}$.

5. $\left(\dfrac{27}{8}\right)^{-2/3} = \left(\dfrac{8}{27}\right)^{2/3} = \left(\left(\dfrac{8}{27}\right)^{1/3}\right)^2 = \left(\dfrac{2}{3}\right)^2 = \boxed{\dfrac{4}{9}}$.

Topic 6 — Rational Equations

1. Multiply by 4: $x + 2 = 5 \Rightarrow \boxed{x = 3}$. ($x = 3$ keeps no denominator 0: valid.)

2. Cross-multiply: $5(x + 3) = 10x \Rightarrow 5x + 15 = 10x \Rightarrow 15 = 5x \Rightarrow \boxed{x = 3}$. (Need $x \neq 0, -3$: valid.)

3. Multiply by $(x-1)$: $2+3(x-1)=5 \Rightarrow 3x-1=5 \Rightarrow 3x=6 \Rightarrow \boxed{x=2}$. (Need $x \neq 1$: valid.)

4. Multiply by x: $12=x^2+4x \Rightarrow x^2+4x-12=0 \Rightarrow (x+6)(x-2)=0$. So $\boxed{x=-6 \text{ or } x=2}$ (both $\neq 0$: valid).

5. Multiply by $(x-3)$: $x=3+4(x-3) \Rightarrow x=4x-9 \Rightarrow x=3$. But $x=3$ makes $x-3=0$, so it is extraneous. **No solution.** (Equivalently: $\frac{x-3}{x-3}=4 \Rightarrow 1=4$, a contradiction.)

Topic 7 — Radical (Root) Equations

1. Square both sides: $x=7^2=\boxed{49}$. (Check: $\sqrt{49}=7$ ✓.)

2. Square both sides: $x-4=6^2=36 \Rightarrow \boxed{x=40}$. (Check: $\sqrt{36}=6$ ✓.)

3. Isolate the radical: $\sqrt{3x+1}=5$; square: $3x+1=25 \Rightarrow 3x=24 \Rightarrow \boxed{x=8}$. (Check: $\sqrt{25}-2=5-2=3$ ✓.)

4. Square: $x+6=x^2 \Rightarrow x^2-x-6=0 \Rightarrow (x-3)(x+2)=0$. Candidates $x=3, -2$. Check: $x=3$ gives $\sqrt{9}=3$ ✓; $x=-2$ gives $\sqrt{4}=2 \neq -2$ **(reject)**. Only $\boxed{x=3}$.

5. Solve $2.5=\sqrt{\dfrac{d}{16}}$. Square both sides: $6.25=\dfrac{d}{16} \Rightarrow d=6.25\times 16=\boxed{100 \text{ feet}}$.